

The background of the page is a photograph of a dark, layered metamorphic rock, possibly a schist or gneiss, showing distinct foliation. A red pen is placed vertically on the left side of the rock to provide a sense of scale. The rock is set against a backdrop of green grass and other rocks.

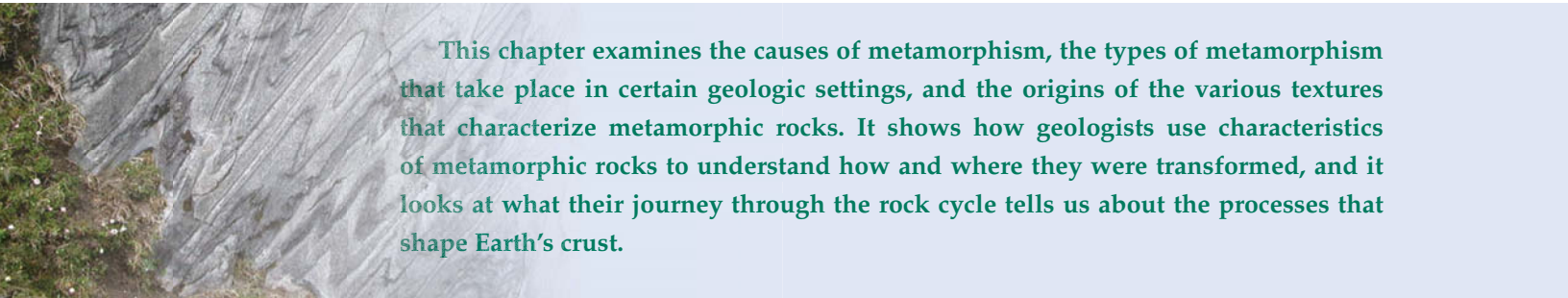
METAMORPHISM: ALTERATION OF ROCKS BY TEMPERATURE AND PRESSURE

6

DURING THE ROCK CYCLE, rocks may be subjected to temperatures and pressures great enough to cause changes in their mineralogy, texture, or chemical composition. We are all familiar with some of the ways in which heat and pressure can transform materials. Cooking batter in a waffle iron not only heats up the batter but also puts pressure on it, transforming it into a rigid solid. In similar ways, rocks are transformed as they encounter high temperatures and pressures deep in Earth's crust.

Tens of kilometers below the surface, temperatures and pressures are high enough to cause chemical reactions and recrystallization that transform rock without being high enough to melt it. Increases in temperature and pressure and changes in the chemical environment can alter the mineral composition and crystalline texture of igneous and sedimentary rock, *even though it remains solid all the while*. The result is the third large class of rocks: metamorphic rocks, which have undergone changes in mineralogy, texture, chemical composition, or all three.

It is important to understand that most metamorphism is a dynamic process, not a static event. Earth's internal heat engine drives the plate tectonic processes that push rocks formed at Earth's surface down to great depths, thereby subjecting them to high pressures as well as high temperatures. But the transformed rocks return to Earth's surface eventually, and that process is largely powered by weathering and erosion—in other words, by the climate system.



This chapter examines the causes of metamorphism, the types of metamorphism that take place in certain geologic settings, and the origins of the various textures that characterize metamorphic rocks. It shows how geologists use characteristics of metamorphic rocks to understand how and where they were transformed, and it looks at what their journey through the rock cycle tells us about the processes that shape Earth's crust.

■ Causes of Metamorphism

Sediments and sedimentary rocks are products of Earth's surface environments, whereas igneous rocks are products of the magmas that originate in the lower crust and mantle. Metamorphic rocks are the products of processes acting on rocks at depths ranging from the upper to the lower crust.

When a rock is subjected to significant changes in temperature or pressure, it will, given enough time—short by geologic standards, but usually a million years or more—undergo changes in its chemical composition, mineralogy, and texture, or all three, until it is in equilibrium with the new temperature and pressure. A limestone filled with fossils, for example, may be transformed into a white marble in which no trace of fossils remains. The mineral and chemical composition of the rock may be unaltered, but its texture may have changed drastically, from small calcite crystals to large, interlocked calcite crystals that erase such former features as fossils. Shale, a well-bedded sedimentary rock so fine-grained that no individual crystal can be seen with the naked eye, may become schist, in which the original bedding is obscured and the texture is dominated by large crystals of mica. In this case, both mineralogy and texture have changed, but the overall chemical composition of the rock has remained the same.

Most metamorphic rocks are formed at depths of 10 to 30 km, in the middle to lower half of the crust. Only later are those rocks *exhumed*, or transported back to Earth's surface, where they may be exposed as outcrops. But metamorphism can also occur at Earth's surface. We can see metamorphic changes, for example, in the baked surfaces of soils and sediments just beneath volcanic lava flows.

The heat and pressure in Earth's interior and its fluid composition are the three principal factors that drive metamorphism. In much of Earth's crust, the temperature increases at a rate of 30°C per kilometer of depth, although that rate varies considerably among different regions, as we will see shortly. Thus, at a depth of 15 km, the temperature will be about 450°C—much higher than the average temperature at Earth's surface, which ranges from 10°C to

20°C in most regions. The contribution of pressure is the result of vertically oriented forces exerted by the weight of overlying rocks as well as horizontally oriented forces developed as the rocks are deformed by plate tectonic processes. The average pressure at a depth of 15 km amounts to about 4000 times the pressure at the surface.

As high as these temperatures and pressures may seem, they are only in the middle range of conditions for metamorphism, as **Figure 6.1** shows. A rock's *metamorphic grade* reflects the temperatures and pressures it was subjected to during metamorphism. We refer to metamorphic rocks formed under the lower temperatures and pressures of shallower crustal regions as *low-grade* metamorphic rocks and those formed under the higher temperatures and pressures at greater depths as *high-grade* metamorphic rocks.

As the grade of metamorphism changes, the assemblages of minerals within metamorphic rocks also change. Some silicate minerals are found mostly in metamorphic rocks: these minerals include kyanite, andalusite, sillimanite, staurolite, garnet, and epidote. Geologists use distinctive textures as well as mineral composition to help guide their studies of metamorphic rocks.

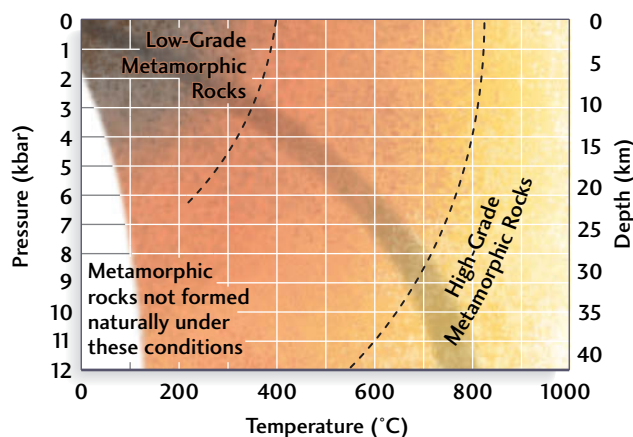


FIGURE 6.1 ■ Temperatures, pressures, and depths at which low-grade and high-grade metamorphic rocks form. The dark band shows the rates at which temperature and pressure increase with depth over much of the continental lithosphere.

The Role of Temperature

Heat can transform a rock's chemical composition, mineralogy, and texture by breaking chemical bonds and altering the existing crystal structures of the rock. When rock is moved from Earth's surface to its interior, where temperatures are higher, the rock adjusts to the new temperature. Its atoms and ions recrystallize, linking up in new arrangements and creating new mineral assemblages. Many new crystals grow larger than the crystals in the original rock.

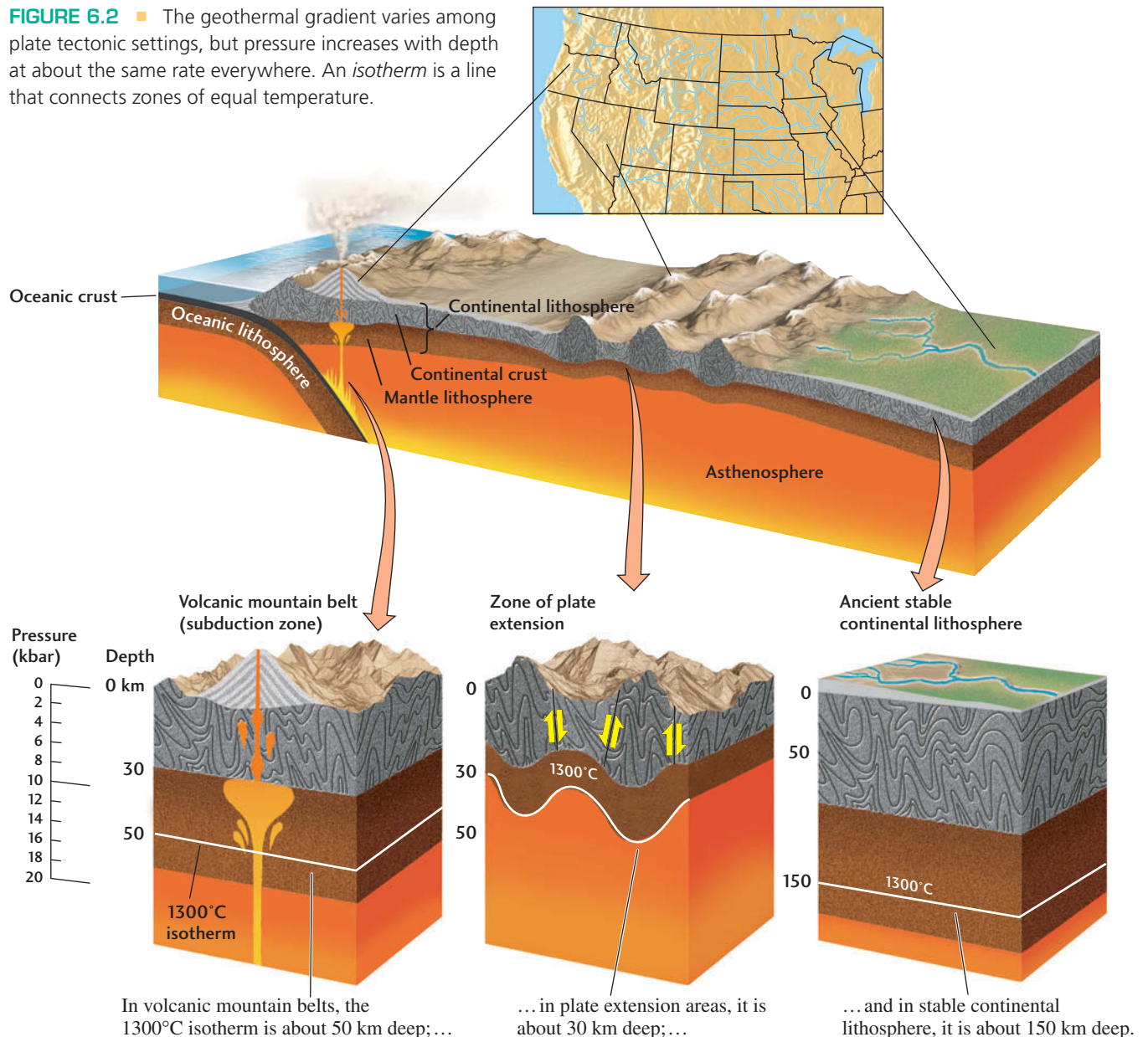
The increase in temperature with increasing depth in Earth's interior is called the *geothermal gradient*. The geothermal gradient varies among plate tectonic settings, but on average it is about 30°C per kilometer of depth. In areas where the continental lithosphere has been stretched and thinned, such as Nevada's Great Basin, the geothermal

gradient is *steep* (for example, 50°C per kilometer of depth). In areas where the continental lithosphere is old and thick, such as central North America, the geothermal gradient is *shallow* (for example, 20°C per kilometer of depth) (Figure 6.2).

Because different minerals crystallize and remain stable at different temperatures, we can use a rock's mineral composition as a kind of *geothermometer* to gauge the temperature at which it formed. For example, as sedimentary rocks containing clay minerals are buried deeper and deeper, the clay minerals begin to recrystallize and form new minerals, such as micas. With additional burial at greater depths and temperatures, the micas become unstable and begin to recrystallize into new minerals, such as garnet.

Plate tectonic processes such as subduction and continent-continent collision, which transport rocks and

FIGURE 6.2 ■ The geothermal gradient varies among plate tectonic settings, but pressure increases with depth at about the same rate everywhere. An *isotherm* is a line that connects zones of equal temperature.



sediments into the hot depths of the crust, are the mechanisms that form most metamorphic rocks. In addition, limited metamorphism may occur where rocks are subjected to elevated temperatures near igneous intrusions. The heat is locally intense, but does not penetrate deeply; thus, the intrusions can metamorphose the surrounding country rock, but the effect is local in extent.

The Role of Pressure

Pressure, like temperature, changes a rock's chemical composition, mineralogy, and texture. Solid rock is subjected to two basic kinds of pressure, also called **stress**:

1. *Confining pressure* is a general force applied equally in all directions, like the pressure a swimmer feels under water. Just as a swimmer feels greater confining pressure when diving to greater depths, a rock descending to greater depths in Earth's interior is subjected to progressively increasing confining pressure in proportion to the weight of the overlying mass.
2. *Directed pressure, or differential stress*, is force exerted in a particular direction, as when you squeeze a ball of clay between your thumb and forefinger. Directed pressure is usually concentrated within particular zones or along discrete planes.

The compressive force exerted where lithospheric plates converge is a form of directed pressure, and it results in deformation of the rocks near the plate boundary. Heat reduces the strength of a rock, so directed pressure is likely to cause severe folding and other forms of ductile deformation, as well as metamorphism, where temperatures are high. Rocks subjected to differential stress may be severely

distorted, becoming flattened in the direction the force is applied and elongated in the direction perpendicular to the force (Figure 6.3).

The minerals in a rock under pressure may be compressed, elongated, or rotated to line up in a particular direction, depending on the kind of stress applied to the rock. Thus, directed pressure guides the shape and orientation of the new crystals formed as minerals recrystallize under the influence of both heat and pressure. During the recrystallization of micas, for example, the crystals grow with the planes of their sheet silicate structures aligned perpendicular to the directed stress. The rock may develop a banded pattern as minerals of different compositions are segregated into separate planes.

Marble owes its remarkable strength to this recrystallization process. When limestone, a sedimentary rock, is heated to the very high temperatures that cause it to recrystallize, the original minerals and crystals become reoriented and tightly interlocked to form a very strong structure with no planes of weakness.

The pressure to which rock is subjected deep in Earth's crust is related to both the thickness and the density of the overlying rocks. Pressure, which is usually recorded in *kilobars* (1000 bars, abbreviated kbar), increases at a rate of 0.3 to 0.4 kbar per kilometer of depth (see Figure 6.1). One bar is approximately equivalent to the pressure of air at Earth's surface. A diver touring the deeper part of a coral reef at a depth of 10 m would experience an additional bar of pressure.

Minerals that are stable at the lower pressures near Earth's surface become unstable and recrystallize into new minerals under the increased pressures deep in Earth's crust. As we will see in Chapter 7, geologists have subjected rocks to extremely high pressures in the laboratory and recorded the pressures required to cause these changes. With these laboratory data



FIGURE 6.3 ■ These rocks in Sequoia National Forest, California, show both the banding and the folding characteristic of sedimentary rocks metamorphosed into marble, schist, and gneiss. [Gregory G. Dimijian/Science Source.]

in hand, we can examine the mineralogy and texture of metamorphic rock samples and infer what the pressures were in the area where they formed. Thus, metamorphic mineral assemblages can be used as pressure gauges, or *geobarometers*. Given a specific assemblage of minerals in a metamorphic rock, we can determine the range of pressures, and therefore the depth, at which the rock must have formed.

The Role of Fluids

Metamorphic processes can alter a rock's mineralogy by introducing or removing chemical components that are soluble in heated water. Hydrothermal fluids accelerate metamorphic chemical reactions because they carry dissolved carbon dioxide as well as other chemical substances—such as sodium, potassium, silica, copper, and zinc—that are soluble in hot water under pressure. As hydrothermal solutions percolate up to the shallower parts of the crust, they react with the rocks they penetrate, changing their chemical and mineral compositions and sometimes completely replacing one mineral with another without changing the rock's texture. This kind of change in a rock's composition by fluid transport of chemical substances into or out of it is called **metasomatism**. Many valuable deposits of copper, zinc, lead, and other metallic ores are

formed by this kind of chemical substitution, as we saw in Chapter 3.

Where do these chemically reactive fluids originate? Although most rocks appear to be completely dry and to have extremely low porosity, they characteristically contain water in minute pores (the spaces between grains). This water comes not from the pores of sedimentary rocks—from which most of the water is expelled during diagenesis—but rather from chemically bound water in clays. Water forms part of the crystal structure of metamorphic minerals such as micas and amphiboles. The carbon dioxide dissolved in hydrothermal fluids is derived largely from sedimentary carbonates: limestones and dolostones.

Types of Metamorphism

Geologists can duplicate metamorphic conditions in the laboratory and determine the precise combinations of pressure, temperature, and parent rock composition under which particular transformations might take place. But to understand when, where, and how such conditions came about in Earth's interior, we must categorize metamorphic rocks on the basis of their geologic settings (**Figure 6.4**).

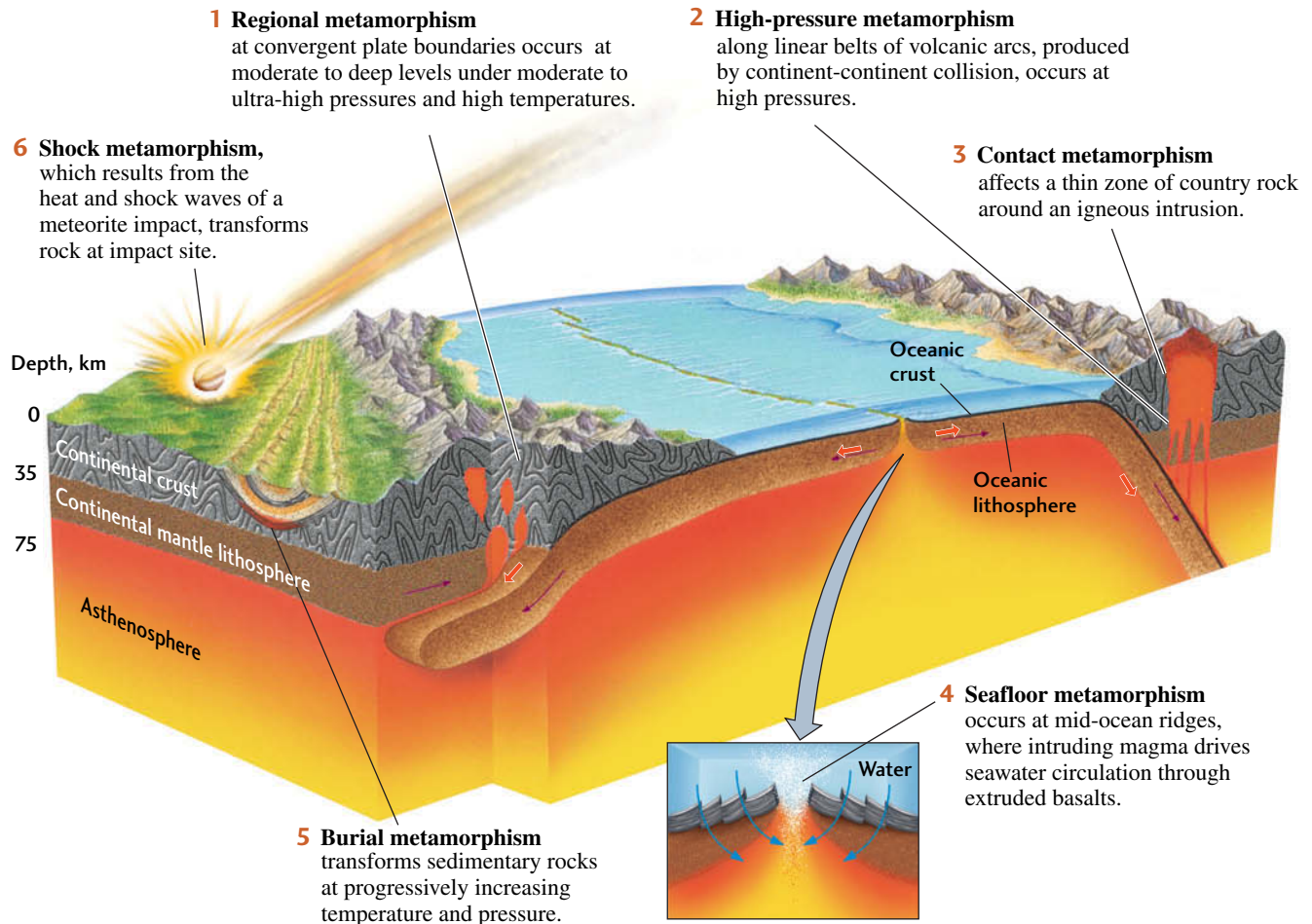


FIGURE 6.4 ■ Different types of metamorphism occur in different geologic settings.

Regional Metamorphism

Regional metamorphism, the most widespread type of metamorphism, takes place where both high temperatures and high pressures are imposed over large parts of the crust. We use this term to distinguish this type of metamorphism from more localized transformations near igneous intrusions or faults. Regional metamorphism is a characteristic feature of convergent plate boundaries. It occurs in volcanic mountain belts, such as the Andes of South America, and in the cores of mountain chains produced by continent-continent collisions, such as the Himalaya of central Asia. These mountain chains are often linear features, so zones of regional metamorphism are often linear in their distribution. In fact, geologists usually interpret regionally extensive belts of metamorphic rocks as representing sites of former mountain chains that were eroded over millions of years, exposing the rocks at their core.

Some regional metamorphic belts are created by the high temperatures and moderate to high pressures near volcanic mountain belts formed where subducted plates sink deep into the mantle. Others are formed under the very high pressures and temperatures found deeper in the crust along boundaries where colliding continents deform rock and raise high mountain chains. In both cases, the metamorphosed rocks are typically transported to great depths in Earth's crust, then eventually uplifted, exposed, and eroded at Earth's surface. A full understanding of the patterns of regional metamorphism, including how rocks respond to systematic changes in temperature and pressure over time, depends on an understanding of the specific plate tectonic settings in which metamorphic rocks form. We will discuss that topic later in this chapter.

into these xenoliths from all directions, and they may become completely metamorphosed.

Contact Metamorphism

In **contact metamorphism**, the heat from an igneous intrusion metamorphoses the rock immediately surrounding it. This type of localized transformation normally affects only a thin zone of country rock along the zone of contact. In many contact metamorphic rocks, especially at the margins of shallow intrusions, the mineral and chemical transformations are largely related to the high temperature of the intruding magma. Pressure effects are important only where the magma is intruded at great depths. Here, the pressure results not from the intrusion forcing its way into the country rock, but from the presence of regional confining pressure. Contact metamorphism by volcanic deposits is limited to very thin zones because lavas cool quickly at Earth's surface and their heat has little time to penetrate the surrounding rocks deeply and cause metamorphic changes. Contact metamorphism may also affect xenoliths that are not completely melted. Blocks of rock up to several meters wide may be torn off the sides of magma chambers and completely surrounded by hot magma. Heat projects

tural variety of metamorphic rock tells us something about the metamorphic process that created it. In this section, we examine those processes, then describe two major textural classes of metamorphic rocks: foliated rocks and granoblastic rocks.

Foliation and Cleavage

The most prominent textural feature of regionally metamorphosed rocks is **foliation**, a set of flat or wavy parallel cleavage planes produced by deformation of igneous and sedimentary rocks under directed pressure (Figure 6.5). These foliation planes may cut through the bedding of the original sedimentary rock at any angle or be parallel to the bedding (Figure 6.5). In general, as the grade of regional metamorphism increases, foliation becomes more pronounced.

A major cause of foliation is the formation of minerals with a platy crystal habit, chiefly the micas and chlorite. The planes of all the platy crystals are aligned parallel to the foliation, an alignment called the *preferred orientation* of the minerals (Figure 6.5). As platy minerals crystallize, their preferred orientation is usually perpendicular to the main direction of the forces squeezing the rock during metamorphism. Crystals of preexisting minerals may contribute to the foliation by rotating until they also lie parallel to the developing foliation plane.

The most familiar form of foliation is seen in slate, a common metamorphic rock, which is easily split into thin sheets along smooth, parallel surfaces. This *slaty cleavage* (not to be confused with the perfect cleavage of sheet silicates such as micas) develops at small, regular intervals in the rock.

Minerals with an elongate, needlelike crystal habit also tend to assume a preferred orientation during metamorphism: these crystals, too, normally line up parallel to the foliation plane. Rocks that contain abundant amphiboles (typically, metamorphosed mafic volcanic rocks) have this kind of texture.

Foliated Rocks

The **foliated rocks** are classified according to four main criteria:

1. Metamorphic grade
2. Grain (crystal) size
3. Type of foliation
4. Banding

Figure 6.6 shows examples of the major types of foliated rocks. In general, foliation progresses from one texture to another with increasing metamorphic grade. In this progression, as temperature and pressure increase, a shale may metamorphose first to a slate, then to a phyllite, then to a schist, then to a gneiss, and finally to a migmatite.

■ Metamorphic Textures

Metamorphism imprints new textures on the rocks it alters. The texture of a metamorphic rock is determined by the sizes, shapes, and arrangement of its constituent crystals. Some metamorphic rock textures depend on the particular kinds of minerals formed under metamorphic conditions. Variation in grain size is also important. In general, grain size increases as metamorphic grade increases. Each tex-

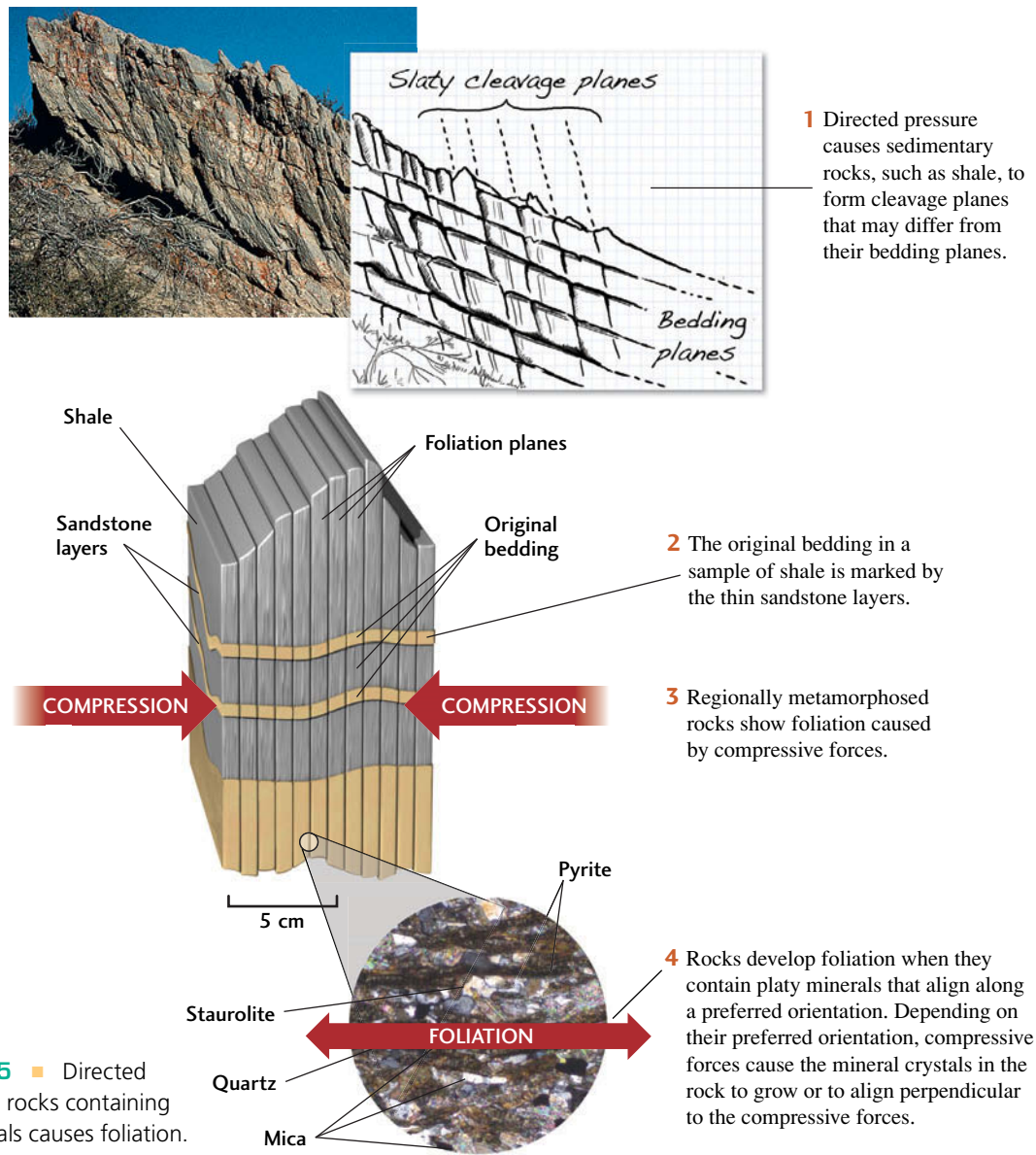


FIGURE 6.5 ■ Directed pressure on rocks containing platy minerals causes foliation.

[(top) Marli Miller; (bottom) John Grotzinger.]

SLATE **Slates** are the lowest grade of foliated rocks. These rocks are so fine-grained that their individual crystals cannot be seen easily without a microscope. They are commonly produced by the metamorphism of shales or, less frequently, of volcanic ash deposits. Slates usually range from dark gray to black, colored by small amounts of organic material originally present in the parent shale. Slate splitters learned long ago to recognize foliation planes and use them to make thick or thin slabs for roofing tiles and blackboards. Flat slabs of slate are still used for flagstone walks in places where slate is abundant.

PHYLLITE **Phyllites** are rocks of a slightly higher grade than the slates, but are similar to them in character and

origin. They tend to have a more or less glossy sheen resulting from crystals of mica and chlorite that have grown a little larger than those of slates. Phyllites, like slates, tend to split into thin sheets, but less perfectly than slates.

SCHIST At low grades of metamorphism, the crystals of platy minerals are generally too small to be seen, and foliation planes are closely spaced. As rocks are subjected to higher temperatures and pressures, however, the platy crystals grow large enough to be visible to the naked eye, and the minerals tend to segregate into lighter and darker bands. This parallel arrangement of platy minerals produces the coarse, wavy foliation known as *schistosity*, which characterizes **schists**. Schists, which are intermediate-grade

As intensity of metamorphism increases, so does crystal size and coarseness of foliation.

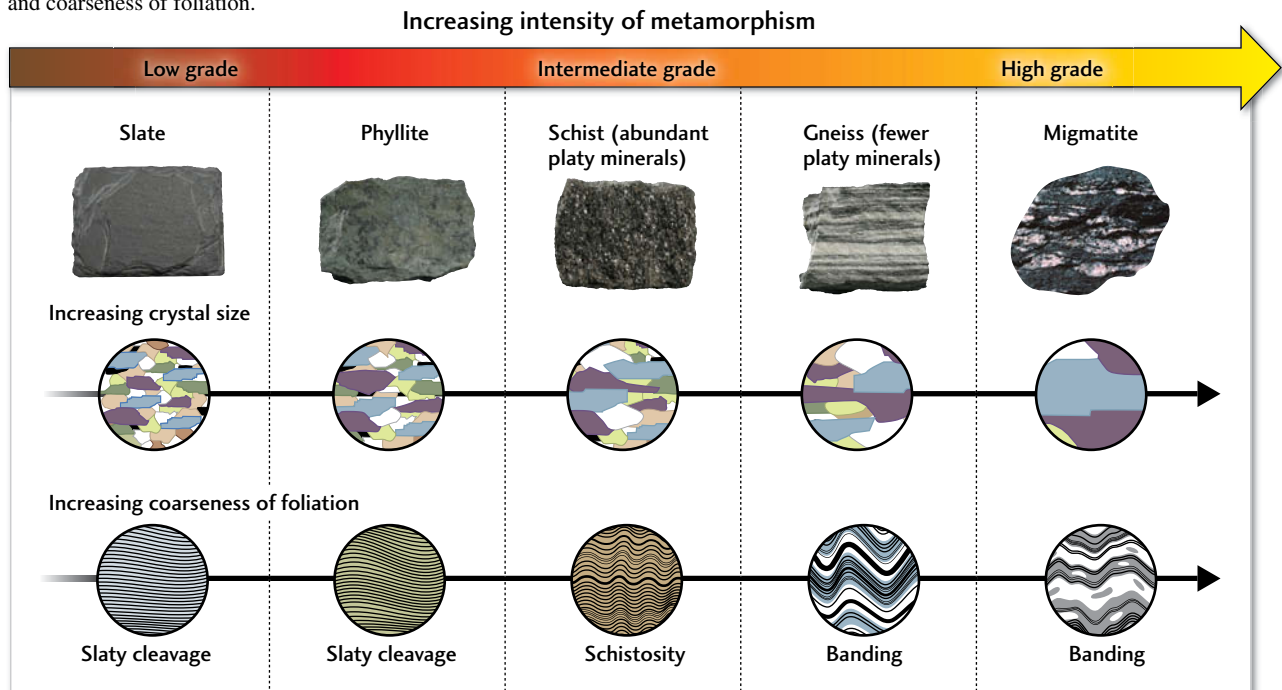


FIGURE 6.6 ■ Foliated rocks are classified by metamorphic grade, grain size, type of foliation, and banding. [slate, phyllite, schist, gneiss: John Grotzinger/Ramón Rivera-Moret/Harvard Mineralogical Museum; migmatite: Kip Hodges.]

rocks, are among the most abundant metamorphic rock types. They contain more than 50 percent platy minerals, mainly the micas muscovite and biotite. Schists may contain thin layers of quartz, feldspar, or both, depending on the quartz content of the parent shale.

GNEISS Even coarser foliation is shown by **gneisses**, light-colored rocks with coarse bands of light and dark minerals throughout the rock. This *gneissic foliation* results from the segregation of lighter-colored quartz and feldspar from darker-colored amphiboles and other mafic minerals. Gneisses are high-grade, coarse-grained metamorphic rocks in which the ratio of granular to platy minerals is higher than that in slate or schist. The result is poor foliation and thus little tendency to split. Under high pressures and temperatures, the mineral assemblages of lower-grade rocks containing micas and chlorite are transformed into new assemblages dominated by quartz and feldspars, with lesser amounts of micas and amphiboles.

MIGMATITE Temperatures higher than those necessary to produce gneiss may begin to melt the country rock. In this case, as with igneous rocks (see Chapter 4), the first minerals to melt will be those with the lowest melting temperatures. Therefore, only part of the country rock melts, and the melt migrates only a short distance before solidify-

ing again. Rocks produced in this way are badly deformed and contorted, and they are penetrated by many veins, small pods, and lenses of melted rock. The result is a mixture of igneous and metamorphic rock called **migmatite**. Some migmatites are mainly metamorphic, with only a small proportion of igneous material. Others have been so affected by melting that they are considered almost entirely igneous.

Granoblastic Rocks

Granoblastic rocks are nonfoliated metamorphic rocks composed mainly of crystals that grow in equant (equidimensional) shapes, such as cubes and spheres, rather than in platy or elongate shapes. These rocks result from metamorphic processes, such as contact metamorphism, in which directed pressure is absent, so foliation does not occur. Granoblastic rocks include hornfels, quartzite, marble, greenstone, amphibolite, and granulite (Figure 6.7). All granoblastic rocks except hornfels are defined by their mineralogy rather than their texture because all of them have a homogeneous granular texture.

Hornfels is a high-temperature contact metamorphic rock of uniform grain size that has undergone little or no deformation (see Figure 3.27). It is formed from fine-grained sedimentary rock and other types of rock containing an



Quartzite



Marble

FIGURE 6.7 ■ Granoblastic (nonfoliated) metamorphic rocks: quartzite [Breck P. Kent]; marble [Diego Lezama Orezza/Corbis].

abundance of silicate minerals. Hornfels has a granular texture overall, even though it commonly contains pyroxene, which makes elongate crystals, and some micas. It is not foliated, and its platy or elongate crystals are oriented randomly.

Quartzites are very hard, white rocks derived from quartz-rich sandstones. Some quartzites are homogeneous, unbroken by preserved bedding or foliation (Figure 6.7a). Others contain thin bands of slate or schist, relics of former interbedded layers of clay or shale.

Marbles are the metamorphic products of heat and pressure acting on limestones and dolomites. Some white, pure marbles, such as the famous Italian Carrara marbles prized by sculptors, show a smooth, even texture of interlocked calcite crystals of uniform size. Other marbles show irregular banding or mottling from silicate and other mineral impurities in the original limestone (Figure 6.7b).

Greenstones are metamorphosed mafic volcanic rocks. Many of these low-grade metamorphic rocks form by sea-floor metamorphism. Large areas of the seafloor are covered with basalts that have been slightly or extensively altered in this way at mid-ocean ridges. An abundance of chlorite gives these rocks their greenish cast.

Amphibolites are made up of amphibole and plagioclase feldspar. They are typically the product of medium- to high-grade metamorphism of mafic volcanic rocks. Foliated amphibolites can be produced by directed pressure.

Granulite, a high-grade metamorphic rock that is also referred to as *granofels*, has a homogeneous granular texture. It is a medium- to coarse-grained rock in which the crystals are equant and show only faint foliation at most. It is formed by the metamorphism of shale, impure sandstone, and many kinds of igneous rock.

Porphyroblasts

Newly formed metamorphic minerals may grow into large crystals surrounded by a much finer grained matrix of other minerals (Figure 6.8). These large crystals, called



FIGURE 6.8 ■ Garnet porphyroblasts in a schist matrix. The minerals in the matrix are continuously recrystallized as pressures and temperatures change and therefore grow to only a small size. In contrast, porphyroblasts grow to a large size because they are stable over a broad range of pressures and temperatures. [MSA 260 by Chip Clark, Smithsonian.]

TABLE 6-1 Classification of Metamorphic Rocks by Texture

Classification	Characteristics	Rock Name	Typical Parent Rock
Foliated	Distinguished by slaty cleavage, schistosity, or gneissic foliation; mineral grains show preferred orientation	Slate Phyllite Schist Gneiss	Shale, sandstone
Granoblastic (nonfoliated)	Granular, characterized by coarse or fine interlocking grains; little or no preferred orientation	Hornfels Quartzite Marble Argillite Greenstone Amphibolite Granulite	Shale, volcanics Quartz-rich sandstone Limestone, dolomite Shale Basalt Shale, basalt Shale, basalt
Porphyroblastic	Large crystals set in fine-grained matrix	Slate to gneiss	Shale

porphyroblasts, are found in rocks formed both by contact and by regional metamorphism. Porphyroblasts form from minerals that are stable over a broad range of pressures and temperatures. Crystals of these minerals grow large while the minerals of the matrix are being continuously recrystallized as pressures and temperatures change, so they replace parts of the matrix. Porphyroblasts vary in size, ranging from a few millimeters to several centimeters in diameter. Garnet and staurolite are two common minerals that form porphyroblasts, although many others are also found. The precise composition and distribution of porphyroblasts of these two minerals can be used to infer the pressures and temperatures that occurred during metamorphism, as we will see later in this chapter.

Table 6.1 summarizes the textural classes of metamorphic rocks and their main characteristics.

Regional Metamorphism and Metamorphic Grade

As we have seen, metamorphic rocks form under a wide range of conditions, and their mineralogies and textures are clues to the pressures and temperatures in the crust where and when they formed. Geologists who study the formation of metamorphic rocks constantly seek to determine the intensity and character of metamorphism more precisely than is indicated by a designation of “low grade” or “high grade.” To make these finer distinctions, geologists “read” minerals as though they were pressure gauges and thermometers. These techniques are best illustrated by their application to regional metamorphism.

Mineral Isograds: Mapping Zones of Change

When we study a broad belt of regional metamorphism, we can see many outcrops, some showing one set of minerals, some showing others. Different zones within the belt may be distinguished by *index minerals*: abundant minerals that each form under a limited range of temperatures and pressures (Figure 6.9). For example, a zone of unmetamorphosed shales may lie next to a zone of weakly metamorphosed slates (Figure 6.9a). As we move from the shale zone into the slate zone, a new mineral—chlorite—appears. Chlorite is an index mineral marking the point at which we move into a new zone with a higher metamorphic grade. If laboratory studies have determined the temperature and pressure at which the index mineral forms, we can draw conclusions about the conditions that existed when the rocks in the zone were formed.

We can use the occurrences of index minerals to make a map of the boundaries between metamorphic zones. Geologists define these boundaries by drawing lines called *isograds* that plot the transitions from one zone to the next. Isograds are used in Figure 6.9a to show a series of mineral assemblages produced by the regional metamorphism of shale in New England. A pattern of isograds tends to follow the deformation features (folds and faults) of a region. An isograd based on a single index mineral, such as the chlorite isograd in Figure 6.9a, provides a good approximate measure of metamorphic pressure and temperature.

To determine metamorphic pressure and temperature more precisely, geologists can examine a group of two or three minerals that have crystallized together. For example, based on laboratory data, a geologist knows that a sillimanite zone that contains orthoclase feldspar and sillimanite must have formed by the reaction of muscovite and quartz

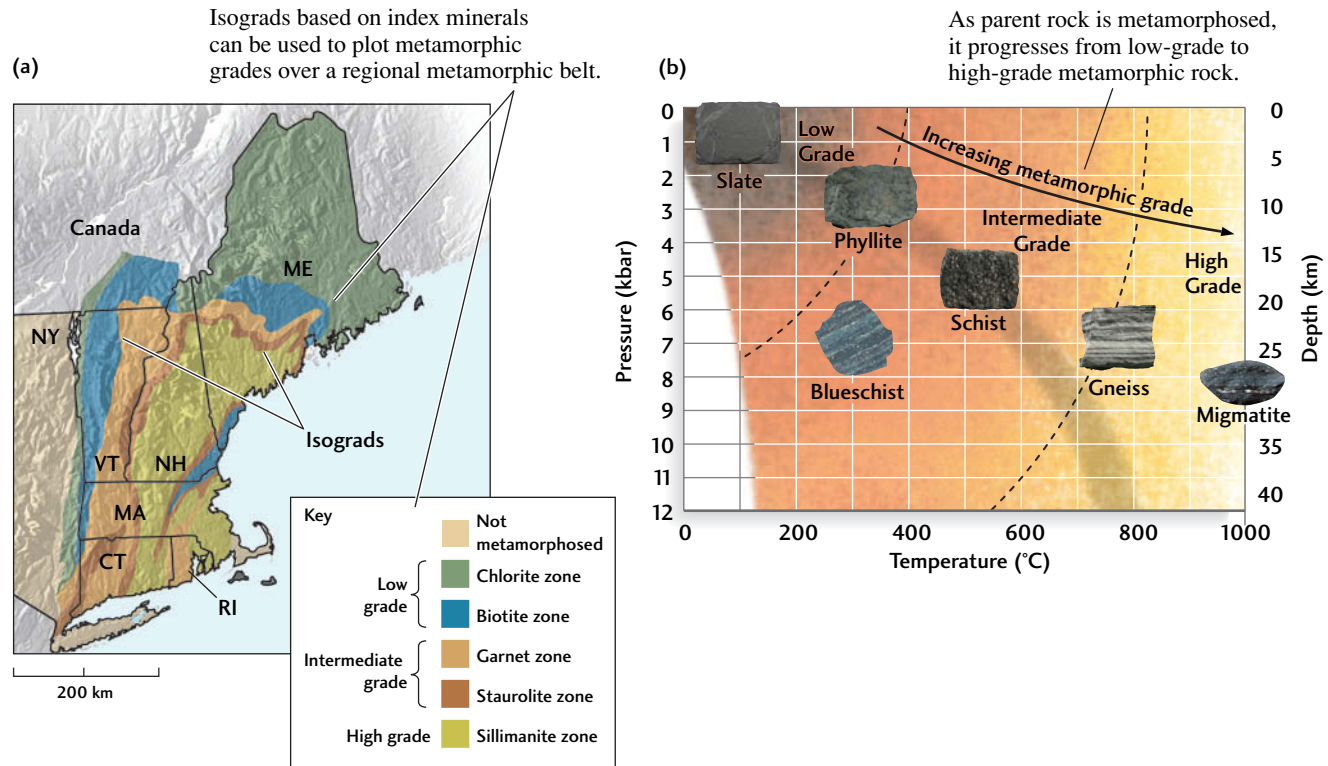
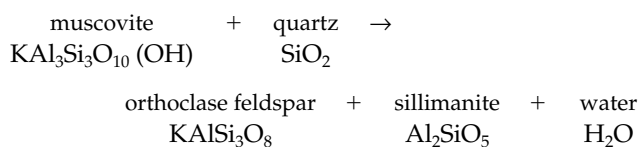


FIGURE 6.9 ■ Index minerals define the different metamorphic zones within a belt of regional metamorphism. (a) Map of New England, showing metamorphic zones based on index minerals found in rocks metamorphosed from shale. (b) Rocks produced by the metamorphism of shale at different temperatures and pressures. [slate, phyllite, schist, gneiss: John Grotzinger/Ramón Rivera-Moret/Harvard Mineralogical Museum; blueschist: courtesy of Mark Cloos; migmatite: courtesy of Kip Hodges.]

at temperatures of about 600°C and pressures of about 5 kbar, liberating water (as water vapor) in the process. The sillimanite isograd records the following reaction:



Because isograds reflect the pressures and temperatures at which the minerals in a regional metamorphic belt formed, the isograd sequence in one belt may differ from that in another. The reason for this variation is that, as we have seen, pressures and temperatures do not increase at the same rate in all plate tectonic settings.

Metamorphic Grade and Parent Rock Composition

The kind of metamorphic rock that results from a given grade of metamorphism depends partly on the mineralogy of the parent rock (Figure 6.10). The metamorphism of shale, as shown in Figure 6.9a, reveals the effects of

pressure and temperature on parent rock that is rich in clay minerals, quartz, and perhaps some carbonate minerals. The metamorphism of mafic volcanic rock, composed predominantly of feldspars and pyroxene, follows a different course (Figure 6.10b).

In the regional metamorphism of a basalt, for example, the lowest-grade rocks characteristically contain various **zeolite** minerals. The silicate minerals in the zeolite class contain water within their crystal structure. Zeolite minerals form at very low temperatures and pressures. Rocks that include this group of minerals are thus placed in the zeolite grade.

Overlapping with the zeolite grade is a higher grade of metamorphosed mafic volcanic rocks, the **greenschists**, whose abundant minerals include chlorite. Next are the amphibolites, which contain large amounts of amphiboles. The granulites, coarse-grained rocks containing pyroxene and calcium-rich plagioclase, constitute the highest grade of metamorphosed mafic volcanic rocks.

Rocks of the greenschist, amphibolite, and granulite grades are also formed during metamorphism of sedimentary rocks such as shale, as shown in Figure 6.10b. The pyroxene-bearing granulites are the products of high-grade metamorphism in which the temperature is high

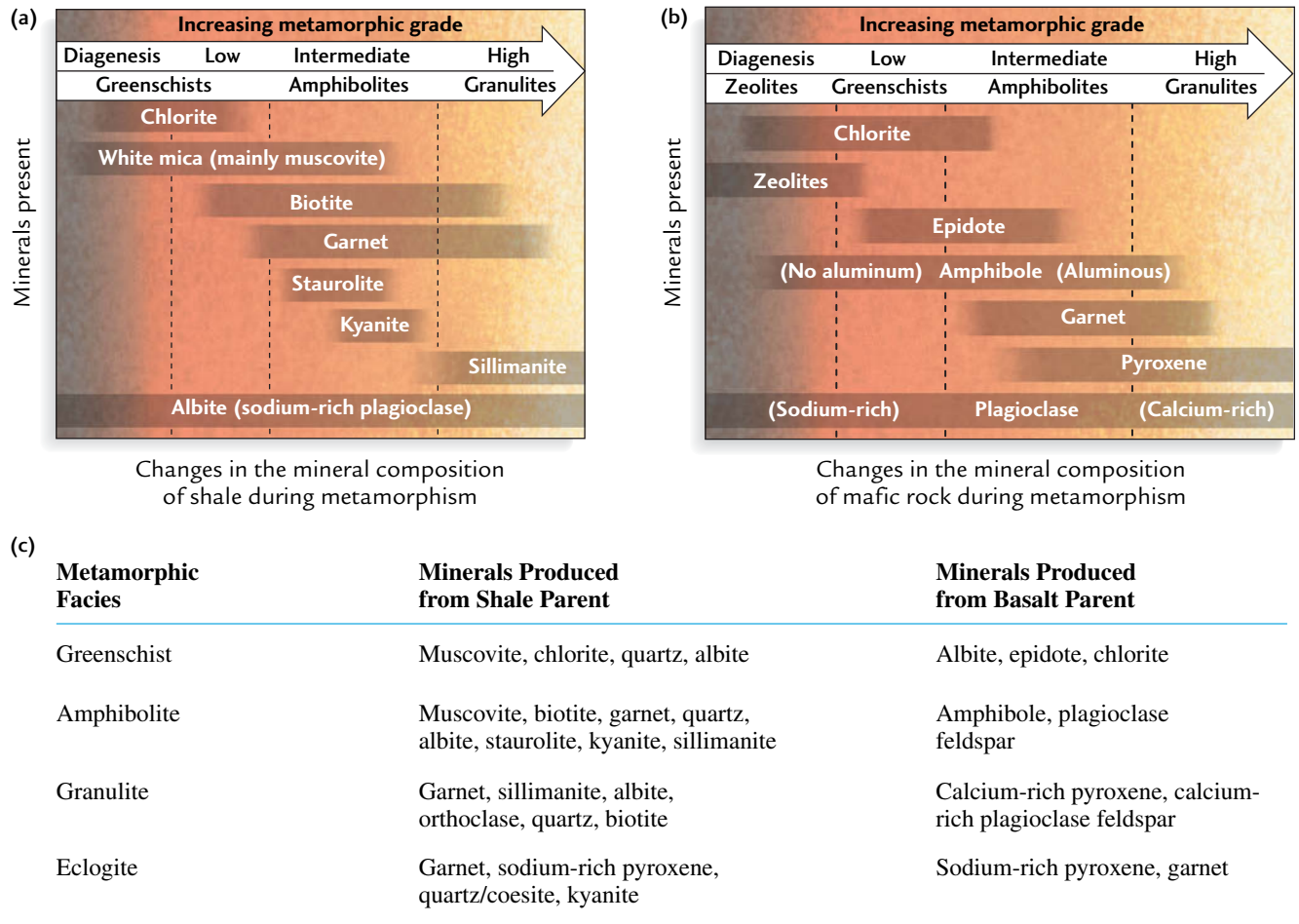


FIGURE 6.10 ■ The kind of metamorphic rock that results from a given grade of metamorphism depends partly on the mineralogy of the parent rock. (a) Changes in the mineral composition of shale (a mafic volcanic rock) with increasing metamorphic grade. (b) Changes in the mineral composition of basalt (a sedimentary rock) with increasing metamorphic grade. (c) Major minerals of metamorphic facies produced from shale and basalt.

and the pressure is moderate. The opposite conditions, in which the pressure is high and the temperature moderate, produce rocks of the **blueschist** grade from parent rock of various starting compositions, from mafic volcanic rocks to shaley sedimentary rocks (see Figure 6.9b). The name comes from the abundance of glaucophane, a blue amphibole, in these rocks. Still another metamorphic rock, formed at extremely high pressures and moderate to high temperatures, is eclogite, which is rich in garnet and pyroxene.

Metamorphic Facies

We can plot this information about the grades of the metamorphic rocks in a regional metamorphic belt—derived from parent rocks of many different chemical compositions—on a graph of temperature and pressure (Figure 6.11). **Metamorphic facies** are groupings of rocks of various

mineral compositions formed under particular conditions of temperature and pressure from different parent rocks. By delineating metamorphic facies, we can be more specific about the grades of metamorphism observed in rocks. Two essential points characterize the concept of metamorphic facies:

1. Different kinds of metamorphic rocks of the same metamorphic grade form from parent rocks of different composition.
2. Different kinds of metamorphic rocks of different metamorphic grades form from parent rocks of the same composition.

Figure 6.10c shows the major minerals of the metamorphic facies produced from shale and basalt. Because parent rocks vary so greatly in composition, there are no sharp boundaries between metamorphic facies (see Figure 6.11).

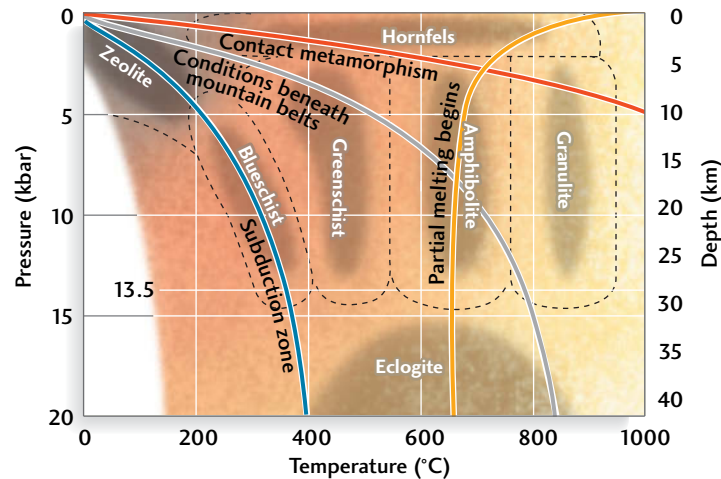


FIGURE 6.11 ■ Metamorphic facies correspond to particular combinations of temperature and pressure, which correspond to particular plate tectonic settings. The dashed lines indicate the overlapping nature of the boundaries between metamorphic facies.

Perhaps the most important reason for analyzing metamorphic facies is that they give us clues to the plate tectonic processes responsible for metamorphism, as we shall see next.

■ Plate Tectonics and Metamorphism

Soon after the theory of plate tectonics was proposed, geologists started to see how patterns of metamorphism fit into the larger framework of plate movements. Different types of metamorphism are likely to occur in different plate tectonic settings (see Figure 6.4):

- *Continental interiors.* Contact metamorphism, burial metamorphism, and perhaps regional metamorphism occur at different levels in the crust. Shock metamorphism is likely to be best preserved in continental interiors because their large areal extent provides a large target area to record rare meteorite impact events.
- *Divergent plate boundaries.* Seafloor metamorphism and contact metamorphism around plutons intruding into the oceanic crust occur at divergent plate boundaries.
- *Convergent plate boundaries.* Regional metamorphism, high-pressure and ultra-high-pressure metamorphism, and contact metamorphism.
- *Transform faults.* In oceanic settings, seafloor metamorphism may occur. In both oceanic and continental settings, we find extensive metamorphism caused by shearing forces along transform faults.

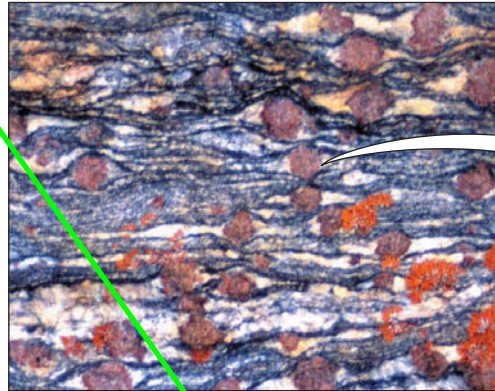
Metamorphic Pressure-Temperature Paths

As we have seen, the concept of metamorphic grade can inform us of the maximum pressure or temperature to which a metamorphic rock has been subjected, but it tells us nothing about where the rock encountered those conditions. Nor does it tell us anything about how the rock was **exhumed**, or transported back to Earth's surface.

Each metamorphic rock has a distinctive history of changing temperature and pressure that is reflected in its texture and mineralogy. This history is called a metamorphic pressure-temperature path, or **P-T path**. The P-T path can be a sensitive recorder of many important factors that influence metamorphism—such as sources of heat, which changes temperatures, and rates of tectonic transport (burial and exhumation), which changes pressures. Thus, P-T paths are characteristic of particular plate tectonic settings.

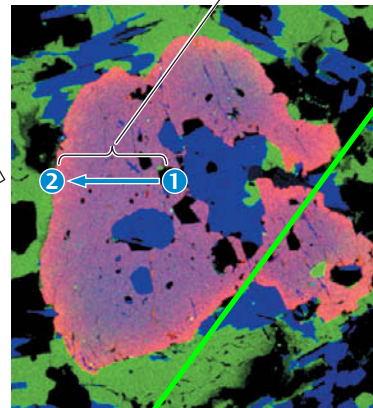
To obtain a P-T path, geologists must analyze specific minerals from metamorphic rock samples in the laboratory. One of the minerals most widely used for this purpose is garnet, a common porphyroblast that serves as a sort of P-T path recording device (Figure 6.12). Garnet crystals grow steadily during metamorphism, and as the pressure and temperature of the environment change, the chemical composition of the garnet changes. The oldest part of a garnet crystal is its core, and the youngest is its outer edge, so the variation in its composition from core to edge will yield the history of the metamorphic conditions under which it formed. Geologists can measure the chemical composition of a garnet porphyroblast in the laboratory and plot the corresponding pressure and temperature values as a P-T path (see Practicing Geology).

- 1 During metamorphism, a garnet crystal grows, and the composition of the growing crystal changes as the temperature and pressure around it change.



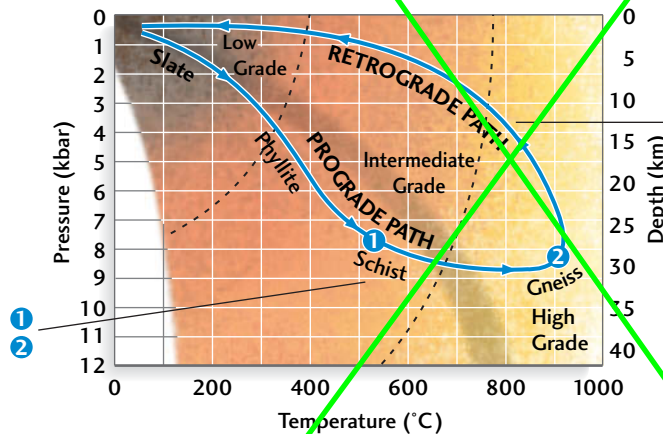
Thin section of garnet gneiss

- 2 The composition of the crystal can be plotted on the P-T path as it grows from 1 in its center to 2 at its edge.



Growth zoning in garnet

- 3 As rock is carried deeper in Earth's crust and is subjected to higher temperatures and pressures (the **prograde** path), the garnet crystal initially grows in a schist but ends up growing in a gneiss as metamorphism progresses.



- 4 The **retrograde** path indicates decreasing temperatures and pressures as rock is carried toward Earth's surface.

FIGURE 6.12 ■ Porphyroblasts such as garnet can be used to plot the P-T paths of metamorphic rocks. The P-T path that a metamorphic rock typically follows begins with an increase in pressure and temperature (the prograde path), followed by a decrease in pressure and temperature (the retrograde path). [Photos courtesy of Kip Hodges.]

P-T paths have two segments: a *prograde* segment that indicates increasing pressure and temperature, and a *retrograde* segment that indicates decreasing pressure and temperature. The P-T paths of some rock assemblages that form at convergent boundaries are shown in [Figure 6.13](#).

Ocean-Continent Convergence

A distinct metamorphic assemblage forms when oceanic lithosphere is subducted beneath a plate carrying a continent on its leading edge (Figure 6.13a). Thick sediments eroded from the continent rapidly fill the deep-sea trench that forms a flexural basin at the subduction zone. As it

descends, the oceanic lithosphere stuffs the region below the inner wall of the trench (the wall closer to the continent) with these sediments, as well as with shreds of ophiolite suites scraped off the descending plate. The result is a chaotic mix known as a **mélange** (French for “mixture”). Assemblages of this type, located in the *forearc* region of a subduction zone—the area between the deep-sea trench and the volcanic mountain belt—are enormously complex and variable. The rocks formed there are all highly folded, intricately faulted, and metamorphosed ([Figure 6.14](#)). They are difficult to map in detail, but are recognizable by their distinctive combination of minerals and structural features.